

The Energy Distribution of Trapped Charges in Polymers Based on Isothermal Surface Potential Decay Model

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ABSTRACT

Space charge formation in polymeric materials can cause some serious concern in real operation, because it has significant influence on the performance of polymers. For example, space charge in some insulating materials can severely distort the electric field, even lead to materials degradation. On the contrary, in the case of its applications, space charge stored in electrets can greatly improve their properties. It is therefore important to understand trapped charge distribution in materials as it is considered to be a novel indicator for effective evaluation of aging status and electric withstanding strength of insulating materials. In this paper, a model based on isothermal surface potential decay (ISPD) is proposed to study the distribution of trapped charges by considering the physical mechanism of the detrapping process. By measuring the ISPD characteristics of polymeric materials and fitting the data according to the assumption of shallow and deep traps, the distribution of trapped charges is obtained, which may be related to the change of aggregation structure of polymers. In order to verify the model, it is used to analyze different ISPD decay curves of polypropylene (PP) and low density polyethylene (LDPE), as well as the ISPD data of PP electrets with/without pressure expanding treatment. The results show that the proposed ISPD model is effective and convenient. Two peaks are observed on the curve of the trapped charge density versus the trap level. The obtained distribution of the trapped charges in polymers can reveal the different nature of electron/hole traps and the different transportation behavior of hole/electron carriers, i.e., the electron-type traps show an inter-chain character while the character of hole-type traps is intra-chain. In addition, the distribution of trapped charge is further related to aggregation structure of PP and LDPE, as well as PP electrets with/without pressure expanding treatment.

Index Terms - Space charge, isothermal surface potential decay (ISPD), trapped charge density, aggregation structure.

1 INTRODUCTION

OVER the years, there has been considerable interest in space charge and charge trap in the research of properties and application of polymeric materials, because the electric field may severely be distorted, leading to part of the material being overstressed even material degradation and possibly premature failure due to the formation of space charge [1-3]. A thoroughly understanding of space charge behavior on the surface and in the bulk of insulating materials is of importance, e.g., improving the performance of diverse applications and furthermore in

reducing safety hazards from or avoiding damage to electronic and electrical equipments [4-5]. Due to the special aggregation structure of polymeric materials, both physical defects in amorphous region and chemical defects like chemical tails of macromolecular chains can lead to the formation of traps. These traps have significant influence on properties of carriers, such as trapping, detrapping, transportation, recombination, etc [6-7]. A lot of work has been done on the research of space charge in polymer films such as low density polyethylene (LDPE) [2, 8], polypropylene (PP) [9], polyimide [10-11], etc. Space charge trapping and detrapping characteristics which are closely related to its formation and the aging process taken place in insulating materials still remain unclear due to the complexity of aggregation structure of polymeric materials [2, 12].

In the past years, many kinds of techniques have been developed to study space charge, such as pulsed electro-acoustic (PEA) [3, 13], laser-induced pressure pulse (LIPP) [14], isothermal surface potential decay (ISPD) [15], etc. ISPD measurement may produce valuable information on charge injection, drift, relaxation, charge traps, etc., provided that the influence of several experimental parameters (as polarity, field, charging technique...) are carefully checked. Moreover, it is expected that ISPD results can be tested and verified by PEA technique, which is another useful and effective space charge testing technique widely adopted in the world [3, 13].

ISPD of insulating materials and photoconductors has been investigated extensively over the past half century. However, the available knowledge is not consistent [15-16]. The classical cross-over phenomena were first observed by Ieda in 1967, which excited theoreticians' imagination and stimulated an upsurge of research on ISPD measurements [17]. And then a series of representative models for ISPD are proposed by many researchers [15]. Ieda and his co-workers considered that the cross-over phenomena may be attributed to two processes of charge injection from the surface into the bulk and charge drift. A detailed theoretical model was given by Batra and his co-workers [18-20]. By assuming field-independent mobility and negligible penetration depth of the surface charge, the effects of partial instantaneous injection and trapping were included in their model. Some theoretical models that include various forms of field-dependent mobility and trapping were developed by Wintle who gave various explanations of the observed characteristics of the polyethylene decay curves [21-22]. Baum et al attributed the rapid decay rate at higher initial potentials to corona-ion-generated light and excited neutral gas molecules of negative charge [23]. Sonnonstine et al gave a convincing ISPD model which includes field-dependent injection efficiency, drift mobility and time-dependent injection, to explain the experimental phenomenon by theoretical calculation [24]. Considering the role of deep traps, Campos et al used the theory proposed by Rudenko to describe the ISPD characteristics of corona-charged insulators [25]. More explanations containing various carrier transport coefficient and charge kinetic equations were considered by Berlepsch [26]. The influence of recombination of electron and hole and energy transfer was taken into account by Vance [27].

According to the results reported in the literature, the charge decay is mostly attributed to surface conduction, dipolar polarization, piezoelectricity, intrinsic conduction, trapping and detrapping by charge carriers surrounding the insulators. However, these factors may work together and it is generally difficult to rule out the dominant one, due to varied conditions such as temperature, relative humidity and charging duration, etc [15]. Detrapping kinetics may become the main factor determining ISPD in the case that transit time of charge carriers is lower than their mean detrapping time especially when the electric field is high. Then we can explain the ISPD in the view of energy [28]. Although some literature interpreted ISPD in the perspective of energy, the classification of the traps and the distinction between electron/hole carriers still need further research.

An ISPD experiment was conducted in Ref [29] and the results were analyzed by a proposed model combining with the IRC theory. The results show that it is effective to measure the energy distribution of electron or hole traps. But some details need further study. In the article, only single trap is taken into account and just one peak is obtained in the computed results of trapped charge distribution. The type of traps is failed to be well distinguished and the presentation of shallow and deep traps is not involved. The effect of trapped charge density on the behavior of polymeric materials is not taken into consideration yet. In addition, a special function $G_n(E, t)$ is defined which is shown as $G_n(E, t) = e_n \exp(-e_n t)$. In order to simplify the calculating process, $G_n(E, t)$ is approximated by the δ function as $G_n(E, t) = (kT/t) \delta(E - E_m)$ in the paper. However, the introduction of δ function may lead to inconsistent units of the two expressions and consequently unreasonable abscissa and ordinate of Figure 4 in Ref [29].

In this paper, an ISPD model was proposed by considering the trapping and detrapping process. The detrapping dynamics is investigated with combination of trap types which are distinguished as shallow traps and deep traps. The trapped charge density versus the trap level can be obtained. Furthermore, the distribution of electrons and holes can be achieved if different charging polarity was selected. The proposed model is used to analyze some ISPD experimental data obtained by corona charged films in order to verify its accuracy and feasibility. Results show that it is convenient and effective to research trapped charges and charge traps, which may imply significant information on aggregation structures of polymeric materials.

2 MODEL OF ISOTHERMAL SURFACE POTENTIAL DECAY

When it comes to model ISPD, the charging process is needed to take into account. The main charging process of an insulating film is by contact (either by a liquid or by a solid), corona discharge and electron beam [15]. The physical system of ISPD technique under consideration usually is that of an insulating sheet, with its free surface charged and its opposing face in good mechanical and electrical contact to an earthed back electrode. Among the charging methods, corona charging is a convenient charging way, and the charging polarity can be varied. Thus it received a wide range of applications. A typical corona charging schematic diagram was shown in Figure 1.

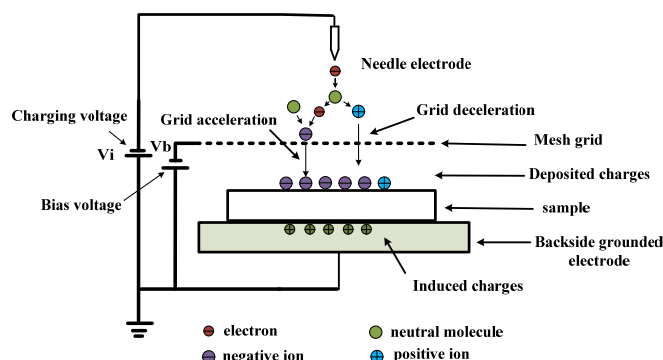


Figure 1. Typical corona charging schematic diagram of ISPD.

The electrons emitted from the needle tip will collide with neutral molecules in the air, which leads to new electrons and positive ions. At the same time, negative ions can be achieved by the combination of electrons and neutral molecules. Taking negatively corona charging as an example, the negative ions and electrons will deposit on the surface of the materials, and the negative carriers (mainly electrons) will penetrate into the material and be captured by different traps on the surface of the film. On the other hand, most of the positive ions will not deposit on the surface due to the deceleration of the grid electrode.

2.1 TIME-DEPENDENT TRAPPED CHARGE DENSITY

The traps may have some discrete energy depths or even continuous distribution due to the complexity of aggregation structure of insulating materials. Several mechanisms, such as photon assisted detrapping, thermal detrapping, impact ionization and tunneling [30], may be responsible for the detrapping process. In the present study, the electric field is not high enough to induce the tunneling or impact ionization and their influence is negligible. The material exposed to visible light with a very low energy is almost impossible to induce photon assisted detrapping. Only thermal detrapping will occur when the charge carriers get enough energy from the lattice thermal vibration. To simplify mathematical calculation, it is assumed that the surface traps are uniformly distributed. In addition, it is assumed that charge carriers will not be retrapped again once it is detrapped due to the high electric field, and the migration process between deep traps and shallow traps is ignored.

Take negatively corona charging as an example, it is assumed that the injected charges are uniformly captured by the surface traps. The detrapping of the trapped charges will form external current when it is short circuited. The detrapping probability of trapped charges is given by [31]

$$P_{de} = \nu_{ATE} \exp\left(-\frac{E_T}{k_B T}\right) \quad (1)$$

where, P_{de} is the detrapping probability of trapped charges, E_T is the energy level of the traps, k_B is the Boltzmann's constant, T is the temperature where the material placed, ν_{ATE} is the attempt-to escape frequency, which can be represented as $\nu_{ATE} = (k_B T)^3 / d h^3 v^2$, here h is the Planck's constant, the value of d is 6, v is the vibration frequency around the defects at the orthogonal plane flow of moving direction [31]. The changing rate of trapped charge density is associated with the trapped charge density and the detrapping probability, which can be expressed as

$$\frac{\partial q_s(t)}{\partial t} = -P_{de} q_s(t) \quad (2)$$

where $q_s(t)$ is the trapped charge density.

By integrating equation (2), we can get the trapped charge density, i.e.,

$$q_s(t) = C_1 \exp\left[-\int_0^t P_{de} dt'\right] \quad (3)$$

The initial conditions can be expressed as

$$q_s(t)|_{t=0} = q_e N_{trap} f(E_T) = q_e N_{trap} \frac{1}{1 + \exp\left(\frac{E_T - E_{Qf}}{kT}\right)} \quad (4)$$

where, N_{trap} is the trap density occupied by carriers at $t=0$, q is the coulomb's quantity of electron, $f(E_T)$ is Fermi Dirac distribution function, E_{Qf} is the Fermi level.

Finally we can obtain the expression of time-dependent trapped charge density as shown in equation (5) by putting equation (4) into equation (3).

$$q_s(t) = q_e N_{trap} f(E_T) \exp\left[-\int_0^t P_{de} dt'\right] \quad (5)$$

2.2 TIME-DEPENDENT EXTERNAL CURRENT

The external current generated by the detrapping of trapped charges can be described as

$$\begin{aligned} j(t) &= \frac{\partial q_s(t)}{\partial t} \\ &= -q_e N_{trap} f(E_T) \nu_{ATE} \exp\left[-\frac{E_T}{k_B T} - \int_0^t \nu_{ATE} \exp\left(-\frac{E_T}{k_B T}\right) dt'\right] \\ &= -q_e N_{trap} f(E_T) G(E_T, t) \end{aligned} \quad (6)$$

here a special function $G(E_T, t)$ is defined and can be expressed as

$$G(E_T, t) = \nu_{ATE} \exp\left[-\frac{E_T}{k_B T} - \int_0^t \nu_{ATE} \exp\left(-\frac{E_T}{k_B T}\right) dt'\right] \quad (7)$$

The function $G(E_T, t)$ is defined to indicate the contribution of trapped electrons to the external current and it can also simplify the computation process [32].

2.3 TRAP LEVEL

The trap level, which is one of the important parameters describing traps, can be achieved by differential computing function $G(E_T, t)$ versus the energy level of traps, i.e.,

$$\frac{\partial G(E_T, t)}{\partial E_T} = 0 \quad (8)$$

Put equation (7) into equation (8) and simplify the equation, we can have

$$\nu_{ATE} t \exp\left(-\frac{E_T}{k_B T}\right) = 1 \quad (9)$$

Finally the expression of energy level can be obtained and it is shown as

$$E_T = k_B T \ln(\nu_{ATE} t) \quad (10)$$

As we can see from equation (10), the energy level is associated with the Boltzmann's constant, Kelvin temperature, attempt-to escape frequency and decay time.

2.4 THE RELATION BETWEEN TRAPPED CHARGES AND POTENTIAL

Considering the change of surface charge density, the electric current density can be achieved and it is given by

$$j(t) = \frac{d\sigma(t)}{dt} \quad (11)$$

where $j(t)$ is the electric current density, $\sigma(t)$ is the surface charge density changing with time.

The surface charge density is described as

$$\sigma(t) = \frac{\epsilon_0 \epsilon_r \phi_s(t)}{L} \quad (12)$$

where ϵ_0 is the vacuum dielectric constant, ϵ_r is relative dielectric constant, $\phi_s(t)$ is the time-dependent surface potential of the material, L is the thickness of the thin film sample.

The relation between the electric current and trapped charge density can be expressed as

$$j(t) = \frac{q_e Q_s(t)}{t} \delta(E_T, t) \quad (13)$$

where $Q_s(t)$ is the trapped charge density and can be shown as $Q_s(t) = q_s(t)/q_e$, $\delta(E_T, t)$ is the Dirac function.

Comprehensively analyze the equation (11), (12) and (13), the trapped charge density is finally obtained, i.e.,

$$Q_s(t) = \frac{j(t) * t}{q_e} = t \frac{\varepsilon_0 \varepsilon_r}{q_e L} \frac{d\phi_s(t)}{dt} \quad (14)$$

As shown in equation (14), the trapped charge density is related to permittivity and thickness of the sample, the decay rate of the ISPD curve and the decay time. The above analysis is also applicable to hole-type traps, and the only difference is that the hole-type trap is below the Fermi level. Thus, the depth of the hole-type trap is referred to the edge of valence band.

To support the model, some data was obtained from certain literature and was discussed based on the ISPD model. The ISPD data is carefully fitted by the MATLAB software according to the assumption of shallow and deep traps. The energy level and the distribution of trapped charges were obtained, and then the dependence of trapped charge density on the energy level can be achieved.

3 RESULTS AND ANALYSIS

3.1 ISPD OF LDPE AND PP

An ISPD experiment of negatively and positively corona charged LDPE and PP with the thickness of 25 μm was conducted in Ref [29]. The sample is first charged using a corona triode and then moved to an electrostatic voltmeter for ISPD measurement. The corona triode consists of a needle, a mesh grid, and a grounded planar electrode. The needle electrode and mesh grid are applied with ± 15 kV and ± 3 kV dc high voltage, respectively. The electric field between the mesh grid and the grounded electrode is kept relatively uniform, and thus, charged particles will uniformly deposit on the sample surface. After charged for 10 min, the sample is moved to the Kelvin probe of electrostatic voltmeter via a rotatable platform. All experiments are carried out in an airtight chamber with the temperature and relative humidity keeping constant as $\sim 40^\circ\text{C}$ and $\sim 40\%$ [29].

Quirke[33-34] *et al.* calculated the distribution of physical and chemical traps in polyethylene (PE) by molecular simulation. The results show that the distribution of trap density versus energy level can be represented by exponential functions. Considering the shallow traps and deep traps, it can be assumed that there are two types of decay processes occurring simultaneously and being controlled by two types of charge. Therefore, a double exponential decay equation can be applied here,

$$V(t) = a \exp(bt) + c \exp(dt) \quad (15)$$

where, $V(t)$ is the surface potential, t is the time, a 、 b 、 c and d are the simulating parameters.

Table 1. The simulating parameters according to equation (15).

Fitting type	LDPE positive	LDPE negative	PP positive	PP negative
<i>a</i>	596.7	-1203	709.7	-675.9
<i>b</i>	-0.001094	-0.007793	-0.001204	-0.001288
<i>c</i>	902.5	-206.1	1130	-1161
<i>d</i>	-6.681e-005	-0.0003938	-6.001e-005	-5.999e-005

The ISPD data was fitted by equation (15) and the corresponding parameters were shown in Table 1. Figure 2 shows the simulation results of PP and LDPE. The simulating curves match the experimental curves well.

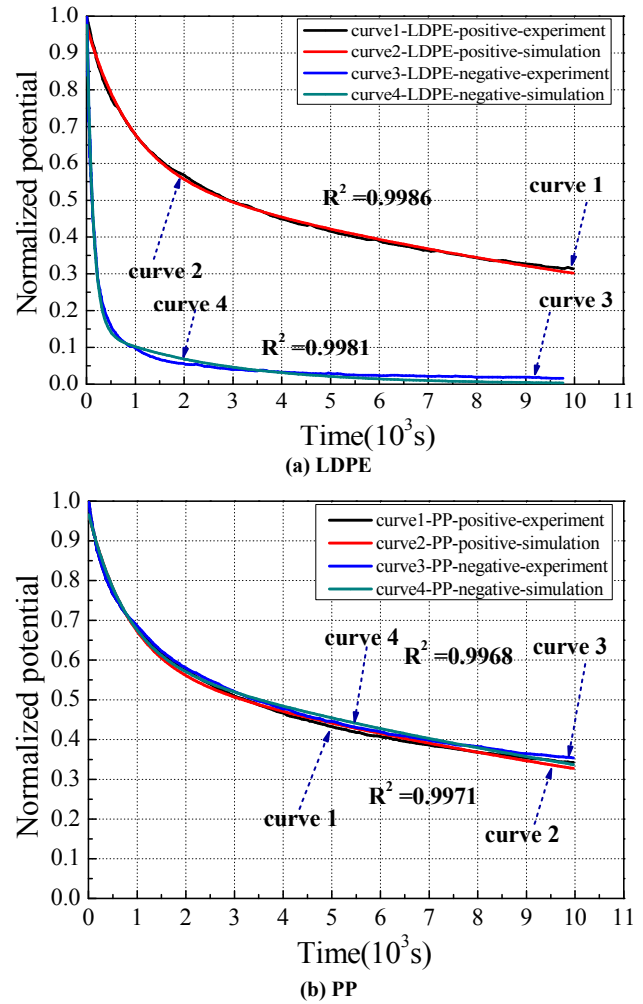


Figure 2. Simulation results of PP and LDPE with positive and negative charging.

According to the model and the simulating parameters, the distribution of trapped charges of negatively and positively charged LDPE and PP are obtained. The results are shown in Figure 3. The insets in the top right of the pictures for comparison are the results from [29]. There are two peaks according to the computed results whether it is a positively or negatively charged LDPE and PP, while the results of the insets exhibit only one peak. The two peaks are consistent with our assumption of shallow and deep traps. Peak 1 represents the maximum value of shallow trapped charge

density and peak 2 represents the deep one. The electron-type trapped charge density is different from hole-type trapped charge density in LDPE while they are almost the same in PP. In LDPE, the peak density of electron-type shallow trapped charges is located at 0.96 eV; while for hole-type ones, it is 1.03 eV with a lower density and much deeper traps. For electron-type and hole-type deep trapped charges, the peak density are located at 1.06 and 1.12 eV, respectively. It is clear that the mean trap depth of hole-type trapped charges is deeper than that of electron-type ones. However, electron-type and hole-type trapped charges have almost the same energy level in PP. The peak density of shallow electron-type and hole-type trapped charges are both located at 1.02 eV and they are both 1.11 eV for deep electron-type and hole-type ones. Additionally, it is found that the deep trapped charge density is higher than the shallow trapped charge density for electron-type and hole-type trapped charges in PP.

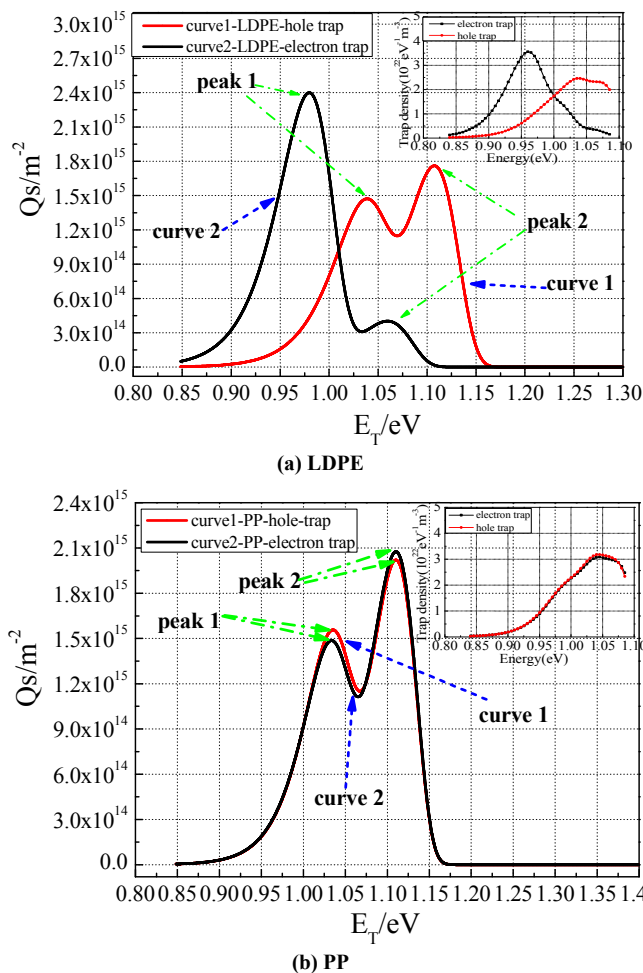


Figure 3. Trapped charge density distribution of LDPE and PP.

The curves in Figure 3 are divided into two single peaks to further analyze the distribution of shallow and deep trapped charges, as exhibited in Figure 4. The shallow and deep trapped charge distribution of electron-type and hole-type traps of LDPE is quite different from that of PP. In LDPE, the energy levels of hole-type shallow and deep trapped charges are located at 0.89-1.08 eV and 0.95-1.16 eV respectively. For electron-type trapped charges, it is 0.85-1.03 eV and 0.93-1.11

eV for shallow and deep trapped charges. Obviously, the density of hole-type deep trapped charges is higher than that of the shallow ones. The maximum value of electron-type shallow trapped charge density is five times higher than that of the deep one, which means that there are mainly electron-type shallow trapped charges in LDPE. For PP, the distribution of electron-type and hole-type shallow trapped charges is almost the same with an energy level of 0.85-1.06 eV. The similar results can be observed in electron-type and hole-type deep trapped charges with an energy level of 0.95-1.17 eV. Figure 4b indicates that electron-type and hole-type trapped charges are mainly deep ones in PP.

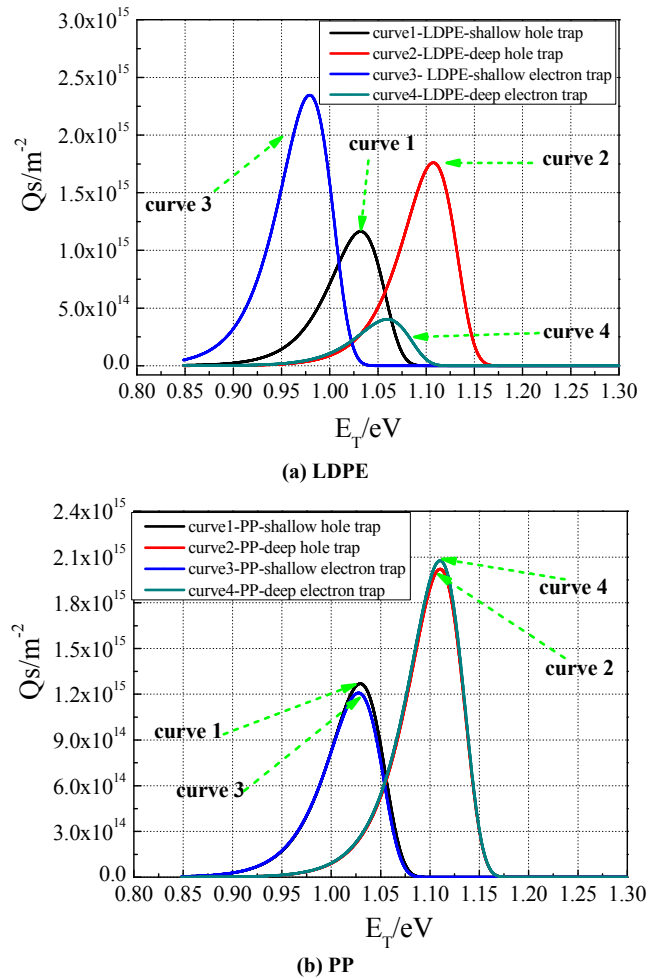


Figure 4. Shallow and deep trapped charge density distribution of LDPE and PP.

3.2 EFFECT OF PRESSURE EXPANDING TREATMENT ON PP ELECTRETS

Over the years, polymeric film electrets have been widely applied to sensors and actuators, energy sources and environmental cleaning engineering, as well as radiation dose measurement, etc. due to their excellent properties such as high stability of polar and non-polar charges, film flexibility, low permittivity and low cost, etc [35]. The feasibility to store charges for a long time in insulating materials was exploited to develop these electret-based devices. Charge decay mechanisms studied here were to improve the time stability of

the charges. Recently, much attention was paid on PP cellular with space charge in the field of the functional dielectric materials. Because it not only has very strong piezoelectricity whose magnitude is similar to or stronger than that of piezoelectric ceramics, but also has excellent space charge storage ability. The properties of PP electrets can be influenced by many factors, such as air humidity and chemical treatment [36-37]. Pressure expanding treatment has a significant influence on the aggregation structure of electrets and then affects their electrical properties [38-39]. More investigation should be carried out on the effects of pressure expanding treatment on electrets applications.

ISPD measurement is a significant method to study important parameters of electrets such as thermal charge stability, trapped charge density etc. A study on PP electrets focusing on the effects of pressure expanding treatment was conducted in Ref [39]. The experimental conditions are strictly controlled. The needle voltage and grid voltage were taken as 9.0, 1.0 and 9.2, 1.2 kV, respectively, to keep the same inner electric field of the samples before and after the expanding due to the rise of the thickness after the expanding treatment. The charging time was set as 5 min. The temperature and relative humidity was kept constant. The decay data of ISPD was used to evaluate the thermal charge stability of PP electrets with/without treatment. The obtained results were satisfying. The ISPD data of PP electrets was obtained and analyzed by our model.

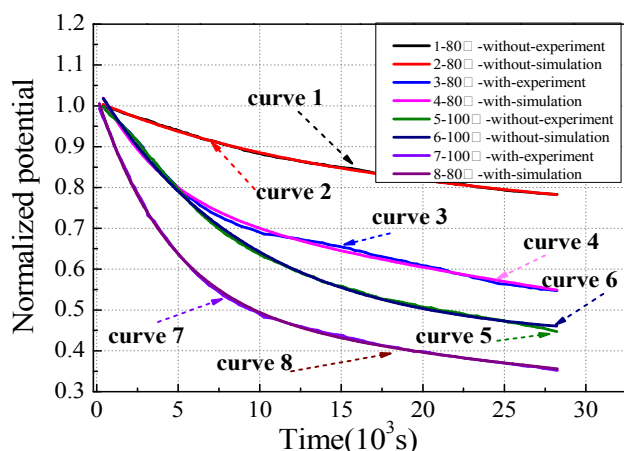


Figure 5. Simulation results of PP electrets with and without expanding treatment.

Figure 5 shows the simulation results of PP electrets with and without treatment. The simulating process and the validation of the accuracy were similar to that in section 3.1. The experimental results and the simulating results are in good conformity. The decay data of ISPD were analyzed according to the model and the simulating parameters. The results are shown in Figure 6.

It can be concluded that the pressure expanding treatment has a significant influence on the distribution of trapped charges. Two peaks related to the maximum value of shallow trapped charge density and deep trapped charge density are observed in each curve. The energy level of the two peaks both become smaller after expanding treatment aged at 80 and 100 °C. After expanding treatment, the peak density of shallow trapped charges increases, while the peak density of deep ones changes

a little. Another remarkable difference between the samples with and without treatment is the range of the energy level. It is clear that the energy level decreases after pressure expanding treatment both aged at 80 and 100°C, which is similar to the phenomenon of spectrum red-shift in spectroscopy.

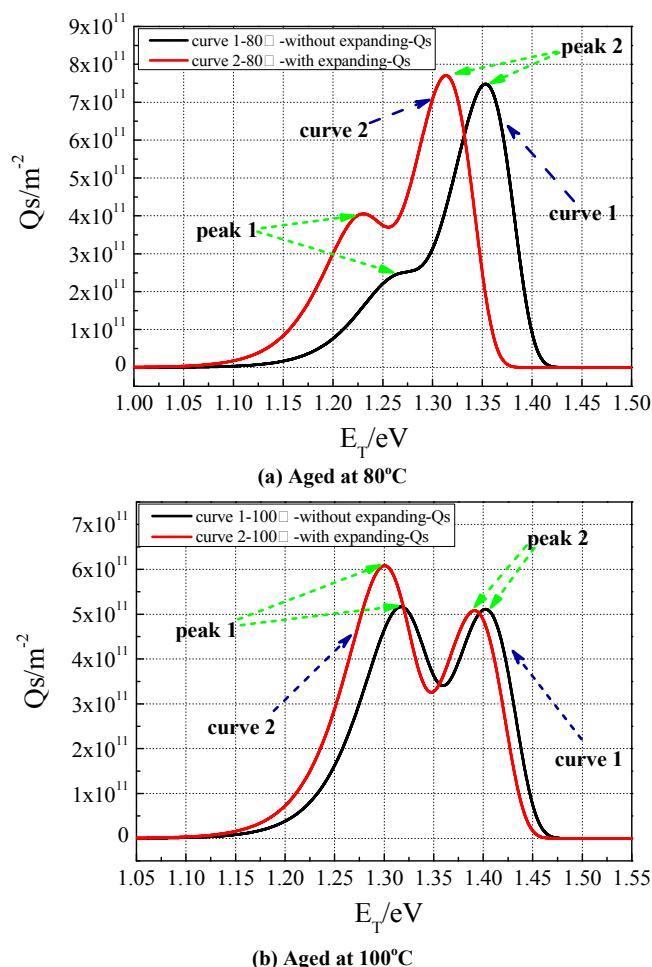


Figure 6. Trapped charge density distribution of PP electrets with/without expanding treatment aged at 80 and 100°C.

The curves are divided into two single peaks to further analyze as shown in Figure 7. A significant change can be observed in the distribution of trapped charges after expanding treatment. For shallow trapped charges, the energy level is 1.14-1.31 eV aged at 80°C and 1.16-1.37 eV aged at 100°C. But the energy level becomes 1.07-1.27 eV and 1.13-1.35 eV after expanding treatment respectively. For deep ones, the energy level changes to 1.17-1.37 eV aged at 80°C and 1.24-1.45 eV aged at 100°C compared to that of 1.20-1.42 eV and 1.25-1.47 eV before expanding treatment. It indicates that the energy level decreases after pressure expanding treatment both for shallow and deep trapped charge density, i.e., pressure expanding treatment may take a great role in reducing the energy level. The maximum shallow trapped charge density aged at 80 and 100°C is enhanced after pressure expanding treatment but the maximum deep trapped charge density only changes a little, which means expanding treatment can cause much more shallow trapped charges and a little more deep trapped charges in PP electrets.

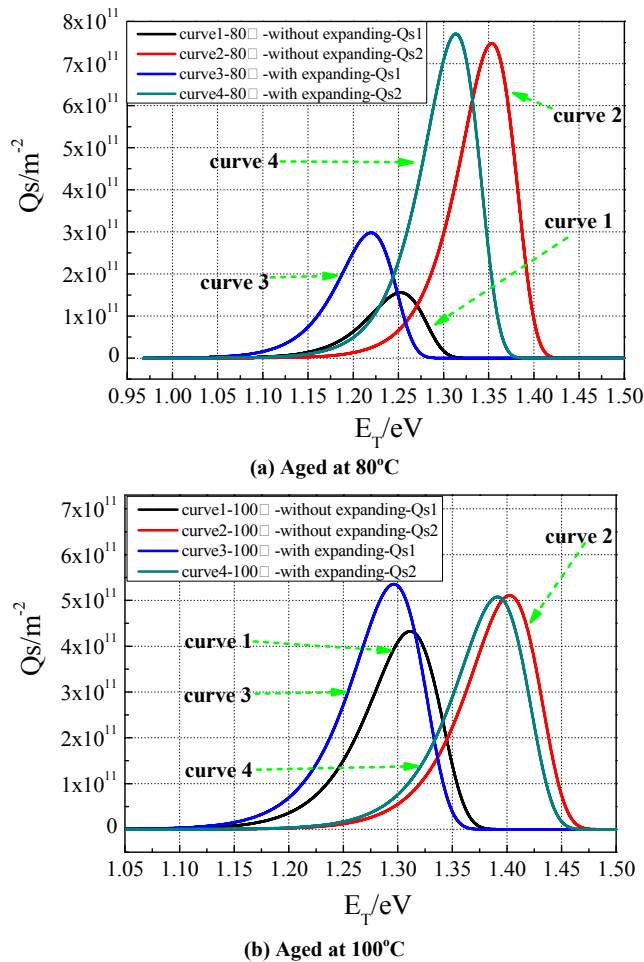


Figure 7. Shallow and deep trapped charge density distribution of PP electrets with/without expanding aged at 80 and 100°C.

In addition, it can be seen from Figure 7 that temperature is another important factor to affect the distribution of trapped charges. The energy level becomes deeper aged at 100 °C than that at 80 °C. At elevated temperature, the shallow trapped charges increases while the deep trapped charges decreases. So the thermal charge stability becomes worse. It can be also found that the quantitative difference between shallow and deep trapped charge density becomes smaller at elevated temperature. In equation (10), the energy levels are positively related with temperature. The attempt-to-escape frequency ν_{ATE} will also increase at a higher temperature. It was suggested that the macroscopic and microscopic parameters of PP electrets can be affected by temperature.

4 DISCUSSION

4.1 ISPD OF LDPE AND PP

The results in Figure 4a and 4b imply some essential difference of aggregation structure of LDPE and PP, which may account for the diversity of electron and hole trapped charge distributions. Generally prepared from a melt phase, PE will consist of ordered crystalline lamellar ribbons to form spherulites and the amorphous phases are formed between the lamellar ribbons [40]. In the amorphous regions, many shorter polymer chains, chain branches and cross-linking sites are

formed and smaller molecules such as residues, antioxidants gases called impurities are collected. The impurities can provide various traps. The interfaces of crystal/amorphous and crystal/vacuum are significant trap staging area and they can provide deep trap energy level of ~1 eV. As shown in [33-34], some chemical defects mainly polar groups such as conjugated double bonds, hydroxyl, carbonyl functions, exist around the chains in the crystalline region. Some chemical defects formed by by-products and additives mainly exist in amorphous region. These chemical defects in crystalline and amorphous regions can contribute to deep traps of ~1 eV in polymeric materials [33-34]. The different mechanisms of electron and hole conduction may play a great role in the distribution of electron and hole trapped charges in LDPE. Some molecular models have been reported to interpret the characteristics of electron and hole conduction [41-42]. The wave function of the conduction-band edge is of inter-chain character, as opposed to the intra-chain character of valence-band states. A conduction electron in its lowest energy state should naturally be channeled out of the crystal and it will prefer to move to a low-density region, such as the outer wall of a cavity, or a crystalline-amorphous interface. They will tend to accumulate where the local PE density is lowest, not highest [41-43]. In other words, an added electron will mainly be repelled by macromolecular chains and trapped in the amorphous region where many shallow traps are formed; while as to an added hole, it will mainly belong to the chain backbone due to the chemical defects with strong bonding force. So, in LDPE, the hole-type traps are relatively deeper than electron-type ones, and more shallow electron-type trapped charges and deep hole-type trapped charges are found.

When it comes to PP, the energy level of electron-type charges is deeper than that in LDPE. It is proposed that the change of interface characteristic in PP may be responsible for this. PP is a linear macromolecule produced by dissociating double bonds between two carbon atoms of asymmetrical propylene monomer. In the case of PP, different configurations (tacticity) of the methyl groups along the main chain skeleton are possible due to asymmetric propylene monomer [44]. Generally speaking, amorphous phases of PE and PP have the same density of ~0.85 g/cm³, but the density (~0.95 g/cm³) of PP crystalline phase is lower than that (~1.0 g/cm³) of PE crystalline phase, due to the presence of methyl in each constitutional unit of PP. It may thus be indicated that crystalline phase of PP is relatively looser or possess more branching chains, which results in different interface characteristics of crystalline region and amorphous region. It is generally accepted that more deep traps can be found in the interface of crystal/amorphous [29]. The different characteristics of crystalline/amorphous interface in LDPE and PP may contribute to the different distribution of electron-type trapped charges. On the other side, the interface of electrode/dielectric likely plays a very important role in charge injection or charge exchange process [8, 45]. As mentioned by Chen [8], the charging current will be varied due to different interfaces, and it is reasonable to suggest that the charge injection or charge exchange process be closely associated with the interfaces. Since the experiment of corona charging

was conducted in air, the carriers were injected into the materials from the interfaces of air/PP or air/LDPE. Thus, different charge accumulation characteristics in LDPE and PP were partly supposed to be caused by air/PP and air/LDPE interfaces. The analysis of chemical defects in PP which produce most of its hole trap states is similar to that in LDPE. In addition, the morphology of polymers can be affected by many factors, such as heat treatment, nucleating agents etc [44, 46-47], which can result in different morphologic properties. Therefore, different density and crystallinity of polymers may be obtained, which will consequently lead to varied charge accumulation performance. Furthermore, due to the complexity of polymer structure, the roles of trapped charge and morphology on trap distribution need more detailed research in future.

4.2 EFFECT OF PRESSURE EXPANDING TREATMENT ON PP ELECTRETS

As mentioned in Ref [39], the pressure expanding treatment can significantly affect the thermal charge stability and trapped charge density of PP electrets. The curve of the sample with pressure expanding treatment shows a faster decay rate than that of the untreated ones, which indicates that the pressure expanding treatment leads to worse thermal charge stability. The amount of trapped charges can be obtained by integrating the curves in Figure 6 in the whole range of the energy level. After pressure expanding treatment, the amount of trapped charges is larger than that of the untreated ones both aged at 80 and 100°C. Furthermore, the model can provide more information of trapped charge distribution and trap level, which is effective to study the change of aggregation structure of polymers with/without treatment.

The change of the morphological characters may be responsible for the results of the samples without/with expanding treatment. The aggregation structure was seriously affected by the pressure expanding treatment and some significantly changes can be obtained. Similar to PE, PP usually contains ordered crystalline lamellar ribbons to form spherulites and amorphous regions. The amorphous phase is an active system in which the chain segments are continuously undergoing thermally-induced conformational changes. The conformational changes are widely characterized by three relaxation processes in which the process containing the propagation twists is particularly important. It controls the movement of tie molecules between lamellae and thereby influences factors such as free-volume and incipient void generation and the break-up of lamellae [40]. All the relaxations are well considered as key parameters in the time-dependent ductile or brittle fracture of polymer under stress [34,40,48]. If the polymer is under pressure expanding treatment, the amorphous phase rather than the crystallite phase will be the first to undergo significant structural change, being influenced by a combination of relatively weak Van der Waals bonds and the conformations of the chains [34]. Some extension of the ties will occur as they collectively take up the strain and there may be some changes in entanglement. This initial elastic response will be replaced gradually by a ductile one in which individual ties begins to be pulled from the

lamellae. This will lead to increased volume of the amorphous element. Part of the relaxation may involve breaking up and disorganization of lamellae. The fundamental response and process of a polymer under pressure can have significant influence on the amorphous region in which the increase of free-volume and void generation were taken place. As a result, more charge traps were introduced in.

The mechanical drawing in combination with the thermal treatment during the expanding treatment can also affect the crystalline region of semicrystalline polymers such as PE and PP [35]. When a crystalline polymer is drawn, then plastic deformation of crystallites of the polymer generally proceeds from smaller-size and more defective crystallites [38-39]. During the drawn process, tie molecules connecting crystallites are stretched. The crystallites will tilt and be distorted. The twisted chains are separated, even worse broken chains were produced. The constant producing of broken chains can affect the crystallinity of polymers and can provide a large amount of charge traps. A larger interface of crystalline/amorphous phase was generated due to more defective crystallites and the increase of amorphous region during the drawn process. The interface of crystalline/amorphous regions can provide a large range of trap energy level and can act as an important trap accumulation area where many charges can be stored. The effect of mechanical drawing in combination with the thermal treatment during the expanding treatment was magnified comparing to the treatment only containing mechanical drawing.

A larger interface of crystalline/amorphous phase, more defects, a larger amount of broken chains and terminal groups by the expanding may form higher charge density stored in the sample. The shallow trapped charge density increases after the sample is treated, which means that the amorphous region can provide more shallow traps and the results confirm our analysis above. While the deep trapped charge density increases only a little after expanding treatment. This may be due to the change of trap sites of PP cellular. The complicated change of amorphous region and crystalline region of polymers during pressure expanding treatment can change the trap sites or some deep traps can be transformed into shallow ones. The expanding treatment can also lead to the increase of escape frequency, i.e., increase of escape cross-section of detrapped charges [39]. A higher escape frequency and much more shallow traps are responsible for the worse thermal stability of charge storage.

We have to admit that the fitting results and the experimental results are not in full accord. The current we adopted in calculation is produced by the carriers moving along the field direction. However, in the process of the actual experiment, not all the carriers are moving along the electric field direction. The surface conduction produced by the carriers moving along the direction perpendicular to the electric field could become an important factor influencing the surface potential. When the air humidity is high, the influence of the surface conduction will become more obvious. Though the humidity is strictly controlled, some carriers can run off from the surface of the sample surface. The migration process

between deep traps and shallow traps during the detrapping process is ignored. But some scientists consider that charge transport occurs by hopping between localized states in polymeric materials [26, 49-50]. Under high electric field, the influence of retrapping process sometimes can be ignored [51-52]. However, it should be noted that migration process between shallow and deep traps can slightly affect the magnitude of the current. Though there is little difference between the experimental results and fitting results due to above reasons, the ISPD model and the processing procedure were proved effective and accurate.

5 CONCLUSION

In the paper, by considering the detrapping process and physical mechanism of trapped charges, a model of ISPD is proposed. We analyze the ISPD data of PP and LDPE as well as the data of PP electrets with/without pressure expanding treatment combined with the assumption of shallow and deep traps in polymers. Several significant findings are got.

1. The proposed ISPD model is effective to study the trapped charge distribution of electrons/holes and can reflect the release process of trapped charges to some extent.

2. The distribution of hole-type trapped charges is quite different from that of electron-type ones in LDPE. The hole-type traps are relatively deeper than electron-type ones in LDPE. The different properties of electron/hole conduction and the distribution of traps may be responsible for this.

3. In LDPE, holes are generally trapped by chemical defects with strong bonding force, while electrons are mainly trapped in the amorphous region where mainly provides shallow traps. Thus there are more deep hole-type trapped charges and shallow electron-type trapped charges in LDPE as calculated.

4. The electron trap states of PP are much deeper than that in LDPE. This may be attributed to different interface characteristics of LDPE and PP, which results in different charge accumulation characteristics.

5. Pressure expanding combining with thermal treatment can seriously affect the aggregation structure of PP electrets. The energy level decreases after pressure expanding treatment and more charge traps especially shallow traps are introduced in PP electrets.

6. The trapped charge density distribution can reveal the distinct nature of electron and hole traps and then uncover the characteristics of aggregation structure of polymeric materials. On the other hand, the unique aggregation structure of polymeric materials can affect the distribution of charge traps and trapped charges. This model may become a bridge of electrical measurement and study of physical parameters of polymeric aggregation structure.

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